Energy-Efficient Probabilistic Semantic Communication Over Space-Air-Ground Integrated Networks

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***Abstract*—Space-air-ground integrated networks (SAGINs) are emerging as a pivotal element in the evolution of future wireless networks. Despite their potential, the joint design of communication and computation within SAGINs remains a formidable challenge. In this paper, the problem of energy efficiency in SAGIN-enabled probabilistic semantic communication (PSCom) system is investigated. In the considered model, a satellite needs to transmit data to multiple ground terminals (GTs) via an uncrewed aerial vehicle (UAV) acting as a relay. During transmission, the satellite and the UAV can use PSCom technique to compress the transmitting data, while the GTs can automatically recover the missing information. The PSCom is underpinned by shared probabilistic graphs that serve as a common knowledge base among the transceivers, allowing for resource-saving communication at the expense of increased computation resource. Through analysis, the computation overhead function in PSCom is a piecewise function with respect to the semantic compression ratio. Therefore, it is important to make a balance between communication and computation to achieve optimal energy efficiency. The joint communication and computation problem is formulated as an optimization problem aiming to minimize the total communication and computation energy consumption of the network under latency, power, computation capacity, bandwidth, semantic compression ratio, and UAV location constraints. To solve this non-convex non-smooth problem, we propose an iterative algorithm where the closed-form solutions for computation capacity allocation and UAV altitude are obtained at each iteration. Numerical results show the effectiveness of the proposed algorithm.**

***Index Terms*—Space-air-ground integrated network, semantic communication, energy efficiency, computation offloading.**

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I. Introduction

With rapid advancements in satellite and uncrewed aerial vehicle (UAV) technology, the potential of space-air-ground integrated networks (SAGINs) to revolutionize future communication systems is becoming increasingly evident [1], [2], [3]. SAGIN covers satellites, aerial platforms, and terrestrial nodes, making it a multi-level three-dimensional (3D) network [4]. This 3D characteristic gives SAGINs global broadband coverage, which is a major demand in the sixth-generation (6G) wireless networks [5]. Beyond traditional wireless connectivity, 6G is envisioned to provide advanced computing services, enabling intelligent and efficient data processing [6], [7]. Among these innovations, semantic communication has gained significant attention as a paradigm that transcends conventional bit-based transmission [8], [9], [10]. In semantic communication, the transmitter extracts semantic information from raw data before transmission, allowing the receiver to reconstruct the message with minimal data exchange [11]. While this approach enhances communication efficiency by reducing the amount of data transmitted, it also introduces additional computation overhead, which requires a joint communication and computation design [12]. Despite extensive research on satellite and UAV communication, integrated networks across space-air-ground present significant challenges in balancing communication and computation [13], [14], necessitating innovative design strategies.

A. Related Works

1) SAGINs: Existing studies on SAGINs have primarily focused on network architecture and resource management.

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Hierarchical architectures have been explored to enable efficient coordination among space, air, and ground segments in SAGINs. For instance, [15] proposed a flexible, low-latency, and flat SAGIN architecture, demonstrating its advantages in reducing delay and enhancing flexibility through two deployment scenarios. Building on this, [16] introduced software-defined network (SDN) into SAGIN, designing a hierarchical domain-based architecture with a multi-controller deployment strategy to enhance network adaptability. In addition, [17] explored the integration of mobile edge computing (MEC) with SAGIN, examining the system architecture, selection schemes, and handover procedures to combine communication, sensing, and computing.

Resource management in SAGINs is another crucial aspect. [18] developed a scalable task scheduling and resource allocation framework, designed to handle diverse task scenarios and manage heterogeneous task demands efficiently. To improve computing resource utilization in the internet of vehicles (IoV), [19] proposed an edge-cloud architecture based on SDN and network function virtualization, which enhances network flexibility and computing efficiency. Additionally, [20] investigated the use of wireless edge caching to improve quality of service (QoS) and quality of experience (QoE), utilizing relay nodes in SAGINs to pre-cache frequently accessed tasks and reduce transmission delays.

Despite these advances, the aforementioned works all used conventional communication schemes. The integration of semantic communication and SAGINs offers a promising avenue to further enhance efficiency by reducing redundant data transmission while introducing new challenges in computation resource allocation.

2) Semantic Communications: Semantic communication is an emerging paradigm that extends beyond traditional bit transmission by prioritizing the conveyance of message meaning rather than raw data streams [21], [22], [23]. By leveraging advancements in artificial intelligence (AI), it effectively extracts essential information, significantly reducing data size by eliminating redundancies [24], [25], [26]. This approach enhances energy efficiency, spectral efficiency, and overall communication reliability [27], [28].

A fundamental aspect of semantic communication is semantic information representation. In joint source-channel coding (JSCC) schemes, deep neural networks generate semantic representations of transmitted data [29]. Some other approaches utilize knowledge graphs or semantic triplets to structure semantic information [30], [31], [32]. Based on the statistical information of extensive knowledge graphs, probabilistic semantic communication (PSCom) employs probabilistic graphs as knowledge bases to enhance semantic compression [33]. Building upon this, [34] integrated rate-splitting multiple access (RSMA) into PSCom and formulated an energy minimization problem. Incorporating distributed reconfigurable intelligent surfaces (RISs), [35] explored spectral efficiency maximization for PSCom while considering computational constraints.

In PSCom, semantic information is compressed using probabilistic graphs, effectively reducing redundancy. In this context, redundancy refers to messages that can be inferred from the shared probabilistic graph. Furthermore, the computation overhead of PSCom can be mathematically modeled as a piecewise function, thus facilitating the joint design of communication and computation processes.

B. Motivation and Contributions

In SAGINs, communication resources are significantly more constrained than in terrestrial networks due to limited spectrum availability, long propagation delays, etc. Semantic communication offers a promising solution by reducing data transmission requirements through intelligent compression, thereby conserving scarce communication resources. However, this comes at the cost of increased computation overhead, as semantic information extraction demand substantial processing power. The multi-layered structure of SAGINs further complicates this trade-off, as computation tasks must be strategically allocated between satellites and UAVs, each with varying processing capabilities and energy constraints [36]. Effectively integrating PSCom into SAGINs requires a joint optimization of communication and computation, balancing data compression benefits against computational feasibility while ensuring efficient resource allocation across different network layers.

This paper proposes a SAGIN-enabled PSCom framework that combines the benefits of SAGIN and semantic communication. To the best of our knowledge, this is the first work that examines semantic communication resource allocation problem in SAGINs. The key contributions are summarized as follows:

- We propose a SAGIN-enabled PSCom framework in which a satellite transmits data to multiple ground terminals (GTs) via a UAV acting as a relay. During the transmission, the satellite and the UAV can choose to perform semantic compression to reduce the data size of the message using PSCom technique, while the GTs can reconstruct the compressed information using the shared probabilistic graphs. Although this semantic compression can save communication resource, it inevitably incurs additional computation costs. The computation overhead function in PSCom system is modeled as a piecewise function with respect to the semantic compression ratio. Thus, a trade-off between communication and computation is required to achieve optimal performance.
- We formulate a joint communication and computation optimization problem aimed at minimizing the total energy consumption of the system, encompassing both communication and computation components. This is achieved by optimizing key system parameters, including the UAV's location and beamwidth, the allocation of bandwidth, computation capacity, and transmit power for each GT on the UAV, the semantic compression ratio, and the allocation of computation tasks between the satellite and the UAV. The optimization problem is subject to constraints on latency, power budgets, computation capacity, bandwidth availability, semantic compression ratios, and UAV positioning.
- This optimization problem exhibits a pronounced non-convex structure, further complicated by the introduction

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TABLE I
LIST OF MAIN NOTATIONS

Notation	Description
$\mathcal{K}$	Set of GTs, $\mathcal{K} = \{1, 2, \ldots, K\}$
H_{U}	Altitude of the UAV
d_k^{UG}	Distance between UAV and GT k
d^{SU}	Distance between satellite and UAV
r_{SU}	Satellite-to-UAV transmission rate
B_{SU}	Satellite-to-UAV bandwidth
P_{S}	Satellite transmit power
N_0	Noise power spectral density
Θ	Half-power beamwidth of UAV antenna
g_k	Channel gain between UAV and GT k
r_k	UAV-to-GT k transmission rate
b_k	Bandwidth allocated to GT k
p_k	Transmit power allocated to GT k
κ	Computation latency constant
F_{S}	Satellite computation capacity
τ	Computation energy constant
f_k	Computation capacity allocated to GT k at UAV
e_{U}	UAV computation energy
T	Maximum tolerable latency constraint
P_{U}	UAV total power budget
$H_{\mathrm{U}}^{\min}, H_{\mathrm{U}}^{\max}$	UAV altitude limits
B_{U}	UAV total bandwidth
F_{U}	UAV total computation capacity
$\rho_k^{\min}$	Minimum semantic compression ratio for GT k
$\Theta_{\min}, \Theta_{\max}$	UAV antenna beamwidth limits
$a_k^{\mathrm{S}}, a_k^{\mathrm{U}}$	Binary variables for computation task allocation

of the piecewise function that induces non-smooth characteristics. To address this challenging non-convex and non-smooth optimization problem, we propose an iterative algorithm that alternately optimizes six subproblems. At each iteration, closed-form solutions are derived for computation capacity allocation and UAV altitude. Numerical results show the effectiveness of the proposed algorithm.

C. Organization and Notations

The remainder of this paper is organized as follows. Section II introduces the system model, detailing the architecture of the considered SAGIN and the PSCom framework, then formulates the energy efficiency optimization problem. Section III presents the proposed iterative algorithm, including the detailed derivation and algorithm analysis on convergence and complexity. Section IV provides a comprehensive analysis of the simulation results, demonstrating the effectiveness of the proposed algorithm in achieving energy-efficient PSCom over SAGINs. Finally, Section V concludes the paper.

The main notations used in the paper are summarized in Table I.

II. System Model and Problem Formulation

Consider a downlink transmission scenario that a satellite in the space needs to transmit data to multiple GTs on the ground and uses a UAV in the air as a relay, as shown in Fig. 1. The set of GTs is represented by $\mathcal{K} = \{1, 2, \cdots, K\}$,

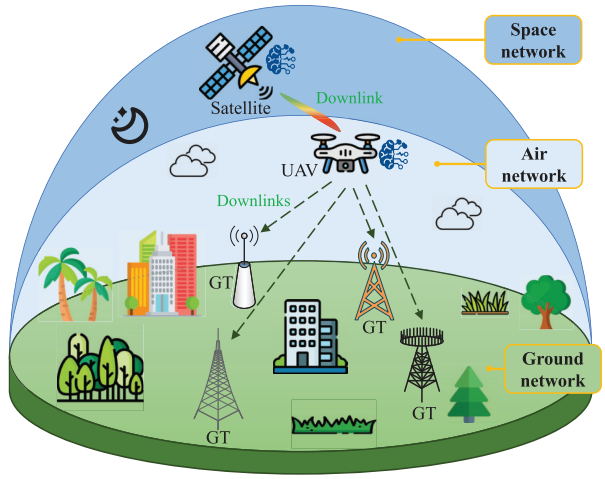

Fig. 1. Illustration of the considered SAGIN-enabled PSCom system.

and the data that need to be transmitted to GT k is denoted by $\mathcal{D}_k$. Due to limited wireless resources, the satellite or UAV needs to take advantage of its computation capability to extract the small-sized semantic information $\mathcal{C}_k$ from original data $\mathcal{D}_k$ to reduce data size, thus saving communication resources. In the considered model, semantic communication is enabled by shared probabilistic graphs between the satellite, UAV, and GTs.

A. Semantic Communication Model

In the considered PSCom model, we assume the transmitter has a set of semantic triplets [32] to transmit. A semantic triplet is defined as

$$\varepsilon = (h, r, t), \tag{1}$$

where h is the head entity, t is the tail entity, and r represents the relation between them. For example, (*Tree, on, Grass*) is a semantic triplet with "*Tree*" as the head entity, "*Grass*" as the tail entity, and "*on*" as the relation. These triplets can be generated from diverse data types (text, images, video) using deep neural networks [37], [38], [39]. As shown, semantic triplets provide high information density, encoding a large amount of data with a small number of bits. Despite their compactness, we propose a probabilistic approach to compress them further.

In traditional semantic triplets, the relation is fixed. However, with a large set of triplets, we can generate a probabilistic graph to store their statistical information. These triplets, derived from historical data between transceivers, form semantic quadruples, which extend the traditional triplet by incorporating the relation probability. The probabilistic graph, as shown in Fig. 2, is shared between transceivers, serving as a knowledge base. The transmitter can then use this graph for semantic compression when sending triplets to the receiver [40]. A multi-dimensional conditional probability matrix is computed from the graph, indicating the likelihood of a specific triplet being valid when others are valid. This allows omitting the relation before transmission, reducing data size. Upon receiving the compressed data, the

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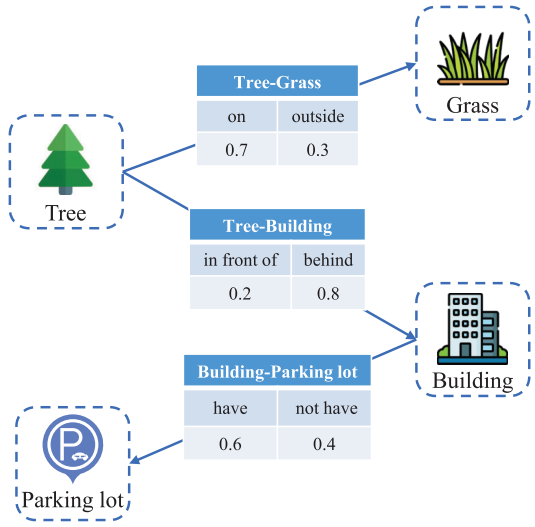

Fig. 2. An illustration of a probabilistic graph in the PSCom system.

receiver can recover the omitted relation using the shared probabilistic graph. For instance, if the transmitter needs to send (*Tree, behind, Building*), it only needs to transmit (*Tree, ∅, Building*), omitting the relation. The receiver can then recover the relation "*behind*" based on the shared graph, as it has a high probability. However, achieving greater data compression requires a lower semantic compression ratio, which involves higher-dimensional conditional probabilities. This leads to increased computation overhead, highlighting the trade-off between communication and computation in the PSCom system.

In the SAGIN-enabled PSCom framework, each GT maintains a local probabilistic graph with statistical data from its history. These graphs are shared with the satellite and UAV. The satellite holds data $\mathcal{D}_k$ for GT k, which may consist of remote sensing results in the form of semantic triplets. Using PSCom, the satellite or UAV can perform semantic compression on $\mathcal{D}_k$. Upon receiving the compressed data $\mathcal{C}_k$, GT k can recover the missing information using the shared graph. The semantic compression ratio for GT k is defined as

$$\rho_k = \frac{C_k}{D_k}, \tag{2}$$

where D_k is the original data size and C_k is the compressed size. A lower ratio indicates more compact compression.

B. Computation Model

The PSCom technique can save communication resources through semantic compression. However, this process inevitably costs additional computation resources.

According to equation (20) in [33], the computation overhead for the considered PSCom technique can be written as

$$O_k(\rho_k) = \begin{cases} A_{k1}\rho_k + B_{k1}, & C_{k1} < \rho_k \le 1, \\ A_{k2}\rho_k + B_{k2}, & C_{k2} < \rho_k \le C_{k1}, \\ \vdots & \\ A_{kD}\rho_k + B_{kD}, & C_{kD} \le \rho_k \le C_{k(D-1)}, \end{cases} \tag{3}$$

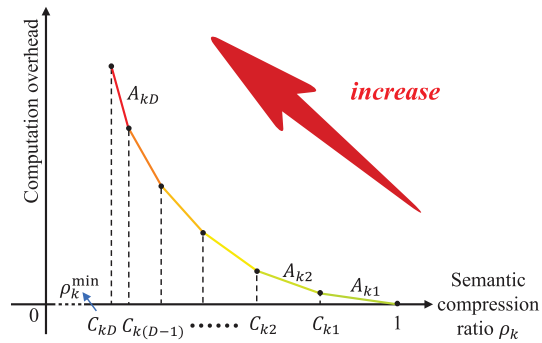

Fig. 3. Illustration of the relationship between semantic compression ratio and computation overhead.

where $A_{kd} < 0$ denotes the slope, $B_{kd} > 0$ represents the constant term, and C_{kd} stands for the boundary for each segment $d \in \mathcal{D}^{\mathrm{s}} = \{1, \cdots, D\}$. These parameters depend on the properties of the probabilistic graph corresponding to different GTs.

From equation (3), we observe that the computation overhead for GT k, denoted by $O_k(\rho_k)$, manifests as a piecewise function with respect to its semantic compression ratio, ρ_k. Fig. 3 visually depicts the behavior of the $O_k(\rho_k)$ function. This function exhibits a segmented structure with D distinct levels, a property that is determined by the dimensionality of the conditional probabilities. This segmentation arises from the inherent characteristic of the semantic compression process, which employs multiple hierarchical levels of conditional probabilities. Each level contributes a distinct linear expression to the overall computation overhead. Furthermore, a discernible decrease in the slope magnitude of $O_k(\rho_k)$ is evident across these discrete segments. This phenomenon can be attributed to the utilization of lower-dimensional conditional probabilities at higher semantic compression ratios, resulting in reduced computation demands. On the contrary, as ρ_k diminishes, the need for higher-dimensional information arises. Consequently, the computation overhead intensifies with increasing information dimensionality. Each transition within the piecewise function $O_k(\rho_k)$ signifies the activation of a higher level of probabilistic information.

C. Network Model

In the considered SAGIN-enabled PSCom system, there is one satellite, one UAV, and K GTs. The satellite and the UAV are equipped with computation capacity and are shared with probabilistic graphs of all GTs. The UAV can hover in the air. Fig. 4 illustrates the considered network.

In the ground network, the horizontal and vertical location of GT k can be denoted by

$$\mathbf{L}_k^{\mathrm{G}} = \left(x_k^{\mathrm{G}}, y_k^{\mathrm{G}}\right), \tag{4}$$

and the height of each GT is approximated to be zero compared to the height of UAV and satellite.

In the air network, the horizontal and vertical location of the UAV can be denoted by

$$\mathbf{L}^{\mathrm{U}} = \left(x^{\mathrm{U}}, y^{\mathrm{U}}\right), \tag{5}$$

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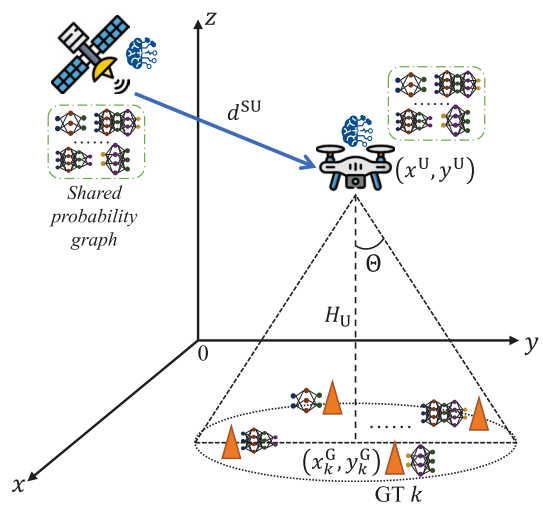

Fig. 4. The considered SAGIN-enabled PSCom network.

and the height of the UAV is represented by H_{U}. Therefore, the distance between the UAV and GT k can be calculated as

$$d_k^{\mathrm{UG}} = \left(\left\| \mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}} \right\|^2 + H_{\mathrm{U}}^2 \right)^{\frac{1}{2}}, \tag{6}$$

where $\|\cdot\|$ is the Euclidian norm.

In the space network, the distance between the satellite and the UAV is denoted as d^{SU}. Given the satellite's significantly high altitude, variations in the UAV's position exert minimal influence on d^{SU}. For analytical simplicity, the impact of the UAV's positional changes on the satellite-UAV distance is therefore neglected. Furthermore, although satellites are inherently mobile, their orbital periods are substantially longer than the transmission timescale under consideration. Consequently, the satellite's orbit is approximated as quasi-static during the data transmission phase, simplifying the analysis.

The satellite transmits data to GT k indirectly with the UAV as a relay. During the transmission process, the satellite or UAV can choose to perform semantic compression using the PSCom technique to reduce the consumption of communication resources. We use a_k^{S} to indicate the computation state of the satellite. If the satellite does the semantic compression for GT k, we have $a_k^{\mathrm{S}} = 1$; otherwise, we have $a_k^{\mathrm{S}} = 0$. Similarly, we use a_k^{U} to indicate the computation state of the UAV. Since the semantic compression for each GT is required at most once, the following constraint can be obtained:

$$a_k^{\mathrm{S}} + a_k^{\mathrm{U}} \leq 1, \tag{7}$$

which indicates that the semantic compression for each GT is either conducted by the satellite, by the UAV, or by neither of them.

D. Transmission Model

In the considered downlink transmission scenario, the satellite sends data to the UAV, which then sends data to the GTs.

1) Satellite to UAV: Different from traditional terrestrial communication, the satellite communication channel is subject to various factors, including space propagation fading, atmospheric absorption fading, rain attenuation, among others. For simplification, we model the satellite-to-UAV wireless channel coefficient as

$$|h_{\mathrm{SU}}| = \frac{\sqrt{\delta_{\mathrm{S}}} \lambda^{\mathrm{SU}}}{4\pi d^{\mathrm{SU}}}, \tag{8}$$

where δ_{S} is the beam gain, and λ^{SU} denotes the wavelength of the satellite-to-UAV transmission wave.

Consequently, the downlink transmission rate between the satellite and the UAV can be given by

$$r_{\mathrm{SU}} = B_{\mathrm{SU}} \log_2 \left(1 + \frac{|h_{\mathrm{SU}}|^2 P_{\mathrm{S}}}{B_{\mathrm{SU}} N_0} \right), \tag{9}$$

where B_{SU} denotes the bandwidth of the satellite-to-UAV system, P_{S} represents the transmit power of the satellite, and N_0 is the power spectral density of additive white Gaussian noise (AWGN).

2) UAV to GT: We assume that the UAV is outfitted with a directional antenna featuring adjustable beamwidth, while each GT is equipped with an omnidirectional antenna possessing unit gain. The azimuth and elevation half-power beamwidths of the UAV's antenna are equivalent and are denoted by $2\Theta \in (0, \pi)$. Conforming to [[41], Eqs. (2)–(52)], the antenna gain within the azimuth angle θ and elevation angle ϕ can be expressed as

$$G = \begin{cases} \dfrac{G_0}{\Theta^2}, & \text{if } 0 \leq \theta \leq \Theta \text{and } 0 \leq \phi \leq \Theta, \\ g, & \text{otherwise}, \end{cases} \tag{10}$$

where $G_0 \approx 2.2846$, and $g \approx 0$ represents the antenna gain outside the beamwidth.

In the considered scenario, the GTs are positioned in outdoor environments, and the communication channels between the UAV and each GT are dominated by the line-of-sight (LoS) path. Consequently, the channel gain between the UAV and GT k can be expressed as[1]

$$g_k = \frac{g_0}{\left(d_k^{\mathrm{UG}}\right)^2}, \tag{11}$$

where g_0 denotes the channel gain at the reference distance of 1 m.

To ensure effective communication between the UAV and the GTs, all GTs must be within the coverage area of the UAV, which means

$$\left\| \mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}} \right\| \leq H_{\mathrm{U}} \tan \Theta, \forall k \in \mathcal{K}. \tag{12}$$

We employ frequency-division multiple access (FDMA) for UAV-to-GT communication, then the achievable downlink transmission rate for GT k satisfying constraint (12) can be written as

$$r_k = b_k \log_2 \left(1 + \frac{G_0 g_k p_k}{\Theta^2 b_k N_0} \right), \tag{13}$$

where b_k is the allocated bandwidth for GT k, and p_k is the allocated transmit power for GT k.

[1]For simplification, we consider the GTs are located outdoors within rural areas where the pathloss exponent is 2, and we assume that communication via sidelobes is negligible.

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E. Latency and Energy Model

Latency and energy efficiency are both important metrics in communication systems, particularly in SAGINs. SAGINs are characterized by exceptionally long communication distances, which require great attention to latency. Additionally, the resource-constrained nature of SAGINs highlights the importance of energy efficiency.

1) Satellite to UAV: The computation latency caused by satellite computation can be modeled as

$$t_{\mathrm{S}} = \frac{\kappa \sum_{k=1}^{K} a_k^{\mathrm{S}} O_k(\rho_k)}{F_{\mathrm{S}}}, \tag{14}$$

where κ is a constant, and F_{S} denotes the computation capacity of the satellite.

Then, the computation energy cost by the satellite can be written as

$$e_{\mathrm{S}} = \tau t_{\mathrm{S}} F_{\mathrm{S}}^3, \tag{15}$$

where τ is a constant.

The communication latency of the satellite-to-UAV link is comprised of transmission delay and propagation delay. The propagation delay cannot be neglected since the vast distance between the satellite and the UAV, often spanning thousands of kilometers.

The transmission delay of the satellite-to-UAV communication can be expressed as

$$t_{\mathrm{T}} = \frac{\sum_{k=1}^{K} \left[a_k^{\mathrm{S}} C_k + \left(1 - a_k^{\mathrm{S}}\right) D_k\right]}{r_{\mathrm{SU}}}, \tag{16}$$

which is data size over transmission rate. The propagation delay of the satellite-to-UAV communication can be expressed as

$$t_{\mathrm{P}} = \frac{d^{\mathrm{SU}}}{c}, \tag{17}$$

where c is the speed of light. Then, the total communication latency of the satellite-to-UAV communication can be written as

$$t_{\mathrm{SU}} = t_{\mathrm{T}} + t_{\mathrm{P}}. \tag{18}$$

Afterwards, the satellite-to-UAV communication energy can be calculated as

$$e_{\mathrm{SU}} = t_{\mathrm{T}} P_{\mathrm{S}}. \tag{19}$$

2) UAV to GT: The computation latency caused by UAV computation for GT k can be modeled as

$$t_k^{\mathrm{U}} = \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{f_k}, \tag{20}$$

where f_k denotes the computation capacity of the UAV that is allocated for GT k.

Then, the total computation energy cost by the UAV can be written as

$$e_{\mathrm{U}} = \tau \sum_{k=1}^{K} t_k^{\mathrm{U}} f_k^3. \tag{21}$$

As the distance between the UAV and each GT is only a few hundred meters, the propagation delay in UAV-to-GT communication can be neglected.

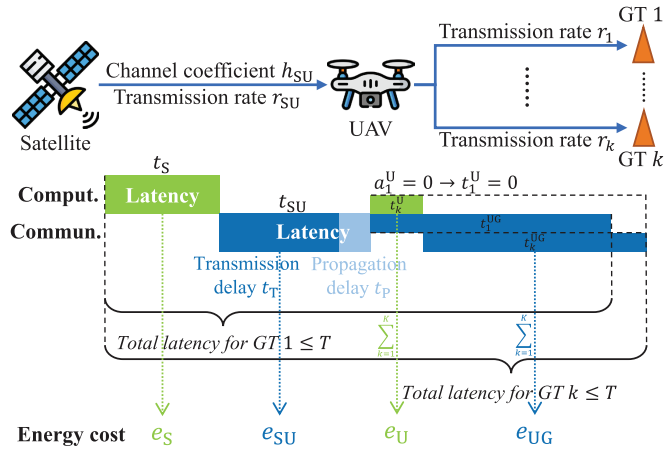

Fig. 5. The framework of the considered SAGIN-enabled PSCom network.

Then, the communication latency from the UAV to GT k that is in coverage can be expressed as

$$t_k^{\mathrm{UG}} = \frac{\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right) \rho_k D_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right) D_k}{r_k}. \tag{22}$$

Afterwards, the total UAV-to-GT communication energy can be calculated as

$$e_{\mathrm{UG}} = \sum_{k=1}^{K} t_k^{\mathrm{UG}} p_k. \tag{23}$$

The general framework of the considered SAGIN-enabled PSCom network is illustrated in Fig. 5.

F. Problem Formulation

Given the defined system model, our goal is to minimize the energy consumption of the SAGIN-enabled PSCom network while considering latency requirement, power budget and suitable location of the UAV, bandwidth and computation capacity allocation of the UAV, semantic compression ratio for each GT, and computation task allocation between the satellite and the UAV. Mathematically, the energy minimization problem can be formulated as

$$\min_{\mathbf{L}^{\mathrm{U}}, H_{\mathrm{U}}, \Theta, \mathbf{b}, \mathbf{f}, \mathbf{p}, \boldsymbol{\rho}, \mathbf{a}^{\mathrm{S}}, \mathbf{a}^{\mathrm{U}}} \quad e_{\mathrm{S}} + e_{\mathrm{SU}} + e_{\mathrm{U}} + e_{\mathrm{UG}}, \tag{24}$$

$$\text{s.t.} \quad t_{\mathrm{S}} + t_{\mathrm{SU}} + t_k^{\mathrm{U}} + t_k^{\mathrm{UG}} \leq T, \forall k \in \mathcal{K}, \tag{24a}$$

$$\sum_{k=1}^{K} p_k \leq P_{\mathrm{U}}, \tag{24b}$$

$$\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\| \leq H_{\mathrm{U}} \tan \Theta, \forall k \in \mathcal{K}, \tag{24c}$$

$$H_{\mathrm{U}}^{\min} \leq H_{\mathrm{U}} \leq H_{\mathrm{U}}^{\max}, \tag{24d}$$

$$\sum_{k=1}^{K} b_k \leq B_{\mathrm{U}}, \tag{24e}$$

$$\sum_{k=1}^{K} f_k \leq F_{\mathrm{U}}, \tag{24f}$$

$$\rho_k^{\min} \leq \rho_k \leq 1, \forall k \in \mathcal{K}, \tag{24g}$$

$$a_k^{\mathrm{S}} + a_k^{\mathrm{U}} \leq 1, \forall k \in \mathcal{K}, \tag{24h}$$

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$$a_k^{\mathrm{S}}, a_k^{\mathrm{U}} \in \{0,1\}, \forall k \in \mathcal{K}, \tag{24i}$$
$$\Theta_{\min} \le \Theta \le \Theta_{\max}, \tag{24j}$$
$$b_k, f_k, p_k \ge 0, \forall k \in \mathcal{K}, \tag{24k}$$

where $\mathbf{b} = [b_1, \cdots, b_K]^{\mathrm{T}}$, $\mathbf{f} = [f_1, \cdots, f_K]^{\mathrm{T}}$, $\mathbf{p} = [p_1, \cdots, p_K]^{\mathrm{T}}$, $\boldsymbol{\rho} = [\rho_1, \cdots, \rho_K]^{\mathrm{T}}$, $\mathbf{a}^{\mathrm{S}} = \left[a_1^{\mathrm{S}}, \cdots, a_K^{\mathrm{S}}\right]^{\mathrm{T}}$, and $\mathbf{a}^{\mathrm{U}} = \left[a_1^{\mathrm{U}}, \cdots, a_K^{\mathrm{U}}\right]^{\mathrm{T}}$. Here, T is the maximum tolerable latency of each GT, P_{U} is the total power budget of the UAV, $\left[H_{\mathrm{U}}^{\min}, H_{\mathrm{U}}^{\max}\right]$ is the feasible altitude range of the UAV, constrained by both obstacle heights and regulatory limitations, B_{U} is the total bandwidth of the UAV, F_{U} is the total computation capacity of the UAV, $\rho_k^{\min}$ is the minimum achievable semantic compression ratio of GT k, and $[\Theta_{\min}, \Theta_{\max}]$ is the feasible range of half-beamwidth for the UAV's antenna, as determined by practical antenna beamwidth tuning techniques.

In problem (24), constraint (24a) requires that the SAGIN-enabled PSCom network cannot have a latency exceeding T for all GTs. This necessitates a careful trade-off between communication and computation, and a strategic allocation of computational tasks between the satellite and the UAV. Constraints (24b), (24e), and (24f) impose limitations on the total power, bandwidth, and computation capacity resources allocated for each GT. Constraints (24c), (24d), and (24j) limit the location of the UAV and the half-beamwidth of its antenna. Constraint (24g) determines the range of the semantic compression ratio of each GT. Constraints (24h) and (24i) govern the allocation of computation tasks between the satellite and the UAV. Finally, constraint (24k) guarantees the non-negativity of bandwidth, computation capacity, and transmit power.

It is generally hard to solve problem (24) since both the objective function and constraint (24a) are non-convex. Additionally, the integer constraint (24i) and the presence of the piecewise function $O_k(\rho_k)$ further complicate the optimization process. To address these challenges and achieve a polynomial-time solution for problem (24), we propose an iterative algorithm that leverages the alternating method.

III. Algorithm Design

In this section, we propose an alternating algorithm to iteratively solve problem (24) by optimizing six subproblems.

A. Satellite-UAV Computation Task Allocation

With given semantic compression ratio, computation capacity, power, and bandwidth allocation, altitude, beamwidth, and location planning, problem (24) can be simplified as

$$\begin{aligned}\min_{\mathbf{a}^{\mathrm{S}},\mathbf{a}^{\mathrm{U}}} \quad & \kappa\tau \sum_{k=1}^{K} a_k^{\mathrm{S}} O_k(\rho_k) F_{\mathrm{S}}^2 \\ & + \frac{P_{\mathrm{S}} \sum_{k=1}^{K} \left[a_k^{\mathrm{S}} \rho_k D_k + \left(1 - a_k^{\mathrm{S}}\right) D_k\right]}{r_{\mathrm{SU}}} \\ & + \kappa\tau \sum_{k=1}^{K} a_k^{\mathrm{U}} O_k(\rho_k) f_k^2 \\ & + \sum_{k=1}^{K} \frac{p_k D_k \left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right) \rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{r_k},\end{aligned} \tag{25}$$

$$\begin{aligned}\text{s.t.} \quad & \frac{\kappa \sum_{k=1}^{K} a_k^{\mathrm{S}} O_k(\rho_k)}{F_{\mathrm{S}}} \\ & + \frac{\sum_{k=1}^{K} \left[a_k^{\mathrm{S}} \rho_k D_k + \left(1 - a_k^{\mathrm{S}}\right) D_k\right]}{r_{\mathrm{SU}}} + \frac{d^{\mathrm{SU}}}{c} \\ & + \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{f_k} \\ & + D_k \frac{\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right) \rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)}{r_k} \le T, \quad \forall k \in \mathcal{K},\end{aligned} \tag{25a}$$
$$a_k^{\mathrm{S}} + a_k^{\mathrm{U}} \le 1, \quad \forall k \in \mathcal{K}, \tag{25b}$$
$$a_k^{\mathrm{S}}, a_k^{\mathrm{U}} \in \{0,1\}, \quad \forall k \in \mathcal{K}. \tag{25c}$$

The difficulty in solving problem (25) arises from the discrete nature of the value space for $\mathbf{a}^{\mathrm{S}}$ and $\mathbf{a}^{\mathrm{U}}$. This characteristic transforms problem (25) into a discrete optimization problem, whose complexity of finding the optimal solution is often significantly high.

To deal with the discrete difficulty of problem (25), we first relax the integer constraint (25b) with

$$a_k^{\mathrm{S}}, a_k^{\mathrm{U}} \in [0,1], \quad \forall k \in \mathcal{K}. \tag{26}$$

Then, problem (25) becomes a convex optimization problem which can be addressed by the dual method.

The dual problem of problem (25) after integer relaxation (26) can be written as

$$\max_{\boldsymbol{\lambda}} \quad D(\boldsymbol{\lambda}), \tag{27}$$

where

$$D(\boldsymbol{\lambda}) = \begin{cases} \min\limits_{\mathbf{a}^{\mathrm{S}},\mathbf{a}^{\mathrm{U}}} & L\left(\mathbf{a}^{\mathrm{S}}, \mathbf{a}^{\mathrm{U}}, \boldsymbol{\lambda}\right), \\ \text{s.t.} & a_k^{\mathrm{S}} + a_k^{\mathrm{U}} \le 1, \forall k \in \mathcal{K}, \\ & a_k^{\mathrm{S}}, a_k^{\mathrm{U}} \in [0,1], \forall k \in \mathcal{K}, \end{cases} \tag{28}$$

with

$$\begin{aligned} L\left(\mathbf{a}^{\mathrm{S}}, \mathbf{a}^{\mathrm{U}}, \boldsymbol{\lambda}\right) = & \kappa\tau \sum_{k=1}^{K} a_k^{\mathrm{S}} O_k(\rho_k) F_{\mathrm{S}}^2 \\ & + \frac{P_{\mathrm{S}} \sum_{k=1}^{K} \left[a_k^{\mathrm{S}} \rho_k D_k + \left(1 - a_k^{\mathrm{S}}\right) D_k\right]}{r_{\mathrm{SU}}} \\ & + \kappa\tau \sum_{k=1}^{K} a_k^{\mathrm{U}} O_k(\rho_k) f_k^2 \\ & + \sum_{k=1}^{K} p_k D_k \frac{\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right) \rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)}{r_k} \\ & + \sum_{k=1}^{K} \lambda_k \left(\frac{\kappa \sum_{k=1}^{K} a_k^{\mathrm{S}} O_k(\rho_k)}{F_{\mathrm{S}}} \right. \\ & + \frac{\sum_{k=1}^{K} \left[a_k^{\mathrm{S}} \rho_k D_k + \left(1 - a_k^{\mathrm{S}}\right) D_k\right]}{r_{\mathrm{SU}}} + \frac{d^{\mathrm{SU}}}{c} \\ & + \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{f_k} \\ & \left. + D_k \frac{\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right) \rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)}{r_k} - T \right), \end{aligned} \tag{29}$$

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and $\boldsymbol{\lambda} = [\lambda_1, \cdots, \lambda_K]$ is non-negative Lagrange multiplier vector with respect to the corresponding constraint (25a).

The objective function in (28) is linear, we can write the coefficient corresponding to a_k^{S} as

$$A_k^{\mathrm{S}} = \kappa\tau O_k(\rho_k)F_{\mathrm{S}}^2 - D_k(1-\rho_k)\left(\frac{P_{\mathrm{S}}}{r_{\mathrm{SU}}} + \frac{p_k}{r_k}\right) + \lambda_k\left[\frac{\kappa O_k(\rho_k)}{F_{\mathrm{S}}} - D_k(1-\rho_k)\left(\frac{1}{r_{\mathrm{SU}}} + \frac{1}{r_k}\right)\right], \quad (30)$$

and write the coefficient corresponding to a_k^{U} as

$$A_k^{\mathrm{U}} = \kappa\tau O_k(\rho_k)f_k^2 - D_k(1-\rho_k)\frac{p_k}{r_k} + \lambda_k\left[\frac{\kappa O_k(\rho_k)}{f_k} - D_k(1-\rho_k)\frac{1}{r_k}\right]. \quad (31)$$

Considering constraint $a_k^{\mathrm{S}} + a_k^{\mathrm{U}} \leq 1$, we can obtain the optimal solution as

$$\left(a_k^{\mathrm{S}}, a_k^{\mathrm{U}}\right)^* = \begin{cases} (0,0), & \text{if } \left(A_k^{\mathrm{S}} \geq 0 \text{ and } A_k^{\mathrm{U}} \geq 0\right), \\ (0,1), & \text{if } \left(A_k^{\mathrm{S}} \geq 0 \text{ and } A_k^{\mathrm{U}} < 0\right) \\ & \quad \text{or } \left(A_k^{\mathrm{U}} < A_k^{\mathrm{S}} < 0\right), \\ (1,0), & \text{if } \left(A_k^{\mathrm{S}} < 0 \text{ and } A_k^{\mathrm{U}} \geq 0\right) \\ & \quad \text{or } \left(A_k^{\mathrm{S}} < A_k^{\mathrm{U}} < 0\right), \end{cases} \quad (32)$$

The value of $\boldsymbol{\lambda}$ can be determined by the sub-gradient method, and the updating process can be given by

$$\lambda_k = \left[\lambda_k + \xi\left(\frac{\kappa\sum_{k=1}^{K} a_k^{\mathrm{S}} O_k(\rho_k)}{F_{\mathrm{S}}} + \frac{\sum_{k=1}^{K}\left[a_k^{\mathrm{S}}\rho_k D_k + \left(1-a_k^{\mathrm{S}}\right)D_k\right]}{r_{\mathrm{SU}}} + \frac{d^{\mathrm{SU}}}{c} + \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{f_k} + D_k\frac{\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1-a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)}{r_k} - T\right)\right]^+, \quad (33)$$

where $[x]^+ = \max\{x, 0\}$, and $\xi > 0$ is the dynamic step size [42]. By iteratively updating $\left(a_k^{\mathrm{S}}, a_k^{\mathrm{U}}\right)$ according to (32) and λ_k according to (33), we can obtain the optimal solution of problem (25) with zero duality gap. Note that although we relaxed the integer constraint (25c) to be continuous, the optimal solution we derived always satisfies the discrete constraint $a_k^{\mathrm{S}}, a_k^{\mathrm{U}} \in \{0,1\}$ in accordance with (32). Therefore, the integer relaxation does not affect the optimality of problem (25).

B. Semantic Compression Ratio Optimization

With given satellite-UAV computation task allocation, computation capacity, power, and bandwidth allocation, altitude, beamwidth, and location planning, problem (24) can be simplified as

$$\min_{\boldsymbol{\rho}} \quad \kappa\tau\sum_{k=1}^{K} a_k^{\mathrm{S}} O_k(\rho_k)F_{\mathrm{S}}^2 + \frac{P_{\mathrm{S}}\sum_{k=1}^{K}\left[a_k^{\mathrm{S}}\rho_k D_k + \left(1-a_k^{\mathrm{S}}\right)D_k\right]}{r_{\mathrm{SU}}} + \kappa\tau\sum_{k=1}^{K} a_k^{\mathrm{U}} O_k(\rho_k) f_k^2 + \sum_{k=1}^{K}\frac{p_k D_k\left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1-a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{r_k}, \quad (34)$$

$$\text{s.t.} \quad \frac{\kappa\sum_{k=1}^{K} a_k^{\mathrm{S}} O_k(\rho_k)}{F_{\mathrm{S}}} + \frac{\sum_{k=1}^{K}\left[a_k^{\mathrm{S}}\rho_k D_k + \left(1-a_k^{\mathrm{S}}\right)D_k\right]}{r_{\mathrm{SU}}} + \frac{d^{\mathrm{SU}}}{c} + \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{f_k} + D_k\frac{\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1-a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)}{r_k} \leq T, \quad \forall k \in \mathcal{K}, \quad (34a)$$

$$\rho_k^{\min} \leq \rho_k \leq 1, \quad \forall k \in \mathcal{K}. \quad (34b)$$

The difficulty in solving problem (34) lies in the piecewise function $O_k(\rho_k)$, which leads to non-smooth optimization.

To address this difficulty, we suggest using the binary variable $\alpha_{kd} \in \{0,1\}$ to signify the linear segment level of $O_k(\rho_k)$. When $\alpha_{kd} = 1$, the computation overhead function $O_k(\rho_k)$ is associated with the d-th segment. Thus, it can be represented as $O_k(\rho_k) = A_{kd}\rho_k + B_{kd}$. Conversely, if $\alpha_{kd} = 0$, the computation overhead function does not pertain to the d-th segment. By introducing the binary variable α_{kd}, we can rewrite the computation overhead function as

$$O_k(\rho_k) = \sum_{d=1}^{D}\alpha_{kd}\left(A_{kd}\rho_k + B_{kd}\right), \quad (35)$$

where D represents the total number of segments of the piecewise function. Furthermore, we have

$$\sum_{d=1}^{D}\alpha_{kd} = 1, \alpha_{kd} \in \{0,1\}, \quad (36)$$

since each GT corresponds to only one linear segment level.

After the above reformulation, we first aim to roughly determine the segment of the piecewise function. To achieve this, we use the midpoint of each segment d to approximate the value of this segment, which can be given by

$$\rho_{kd} = \frac{C_{kd} + C_{k(d-1)}}{2}, \quad \forall d \in \mathcal{D}^{\mathrm{s}}, \ \forall k \in \mathcal{K}, \quad (37)$$

with $C_{k0} = 1, \forall k \in \mathcal{K}$.

Then, the segment selection problem is

$$\min_{\boldsymbol{\alpha}} \quad \kappa\tau F_{\mathrm{S}}^2\sum_{k=1}^{K}\sum_{d=1}^{D} a_k^{\mathrm{S}}\alpha_{kd}\left(A_{kd}\rho_{kd} + B_{kd}\right) + \frac{P_{\mathrm{S}}\sum_{k=1}^{K} D_k\left[a_k^{\mathrm{S}}\left(\sum_{d=1}^{D}\alpha_{kd}\rho_{kd}\right) + \left(1-a_k^{\mathrm{S}}\right)\right]}{r_{\mathrm{SU}}} + \kappa\tau\sum_{k=1}^{K}\sum_{d=1}^{D} a_k^{\mathrm{U}} f_k^2\alpha_{kd}\left(A_{kd}\rho_{kd} + B_{kd}\right) + \sum_{k=1}^{K} p_k D_k\frac{1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}} + \left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\sum_{d=1}^{D}\alpha_{kd}\rho_{kd}}{r_k}, \quad (38)$$

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$$\text{s.t.} \quad \frac{\kappa \sum_{k=1}^{K} \sum_{d=1}^{D} a_k^{\mathrm{S}} \alpha_{kd} \left(A_{kd}\rho_{kd} + B_{kd}\right)}{F_{\mathrm{S}}} + \frac{\sum_{k=1}^{K} D_k \left[a_k^{\mathrm{S}} \left(\sum_{d=1}^{D} \alpha_{kd}\rho_{kd}\right) + 1 - a_k^{\mathrm{S}}\right]}{r_{\mathrm{SU}}} + \frac{d^{\mathrm{SU}}}{c} + \frac{\kappa a_k^{\mathrm{U}} \sum_{d=1}^{D} \alpha_{kd} \left(A_{kd}\rho_{kd} + B_{kd}\right)}{f_k} + D_k \frac{1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}} + \left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right) \sum_{d=1}^{D} \alpha_{kd}\rho_{kd}}{r_k} \leq T, \quad \forall k \in \mathcal{K}, \tag{38a}$$

$$\sum_{d=1}^{D} \alpha_{kd} = 1, \forall k \in \mathcal{K}, \tag{38b}$$

$$\alpha_{kd} \in \{0, 1\}, \forall d \in \mathcal{D}^{\mathrm{s}}, \forall k \in \mathcal{K}. \tag{38c}$$

where $\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1, \cdots, \boldsymbol{\alpha}_k, \cdots, \boldsymbol{\alpha}_K]$ with $\boldsymbol{\alpha}_k = [\alpha_{k1}; \ \cdots; \alpha_{kD}]$.

The optimal solution to problem (38) can be efficiently derived using the dual method, which exhibits a zero duality gap. The procedure closely resembles the steps employed in the first subproblem; therefore, a detailed deduction is omitted here for brevity.

After solving the segment selection problem, we can determine which segment does $O_k(\rho_k)$ belong to. Denote the optimal segment of $O_k(\rho_k)$ by d_k^*, we can rewrite problem (34) as

$$\min_{\boldsymbol{\rho}} \quad \kappa\tau \sum_{k=1}^{K} a_k^{\mathrm{S}} \left(A_{kd_k^*}\rho_k + B_{kd_k^*}\right) F_{\mathrm{S}}^2 + \frac{P_{\mathrm{S}} \sum_{k=1}^{K} \left[a_k^{\mathrm{S}}\rho_k D_k + \left(1 - a_k^{\mathrm{S}}\right) D_k\right]}{r_{\mathrm{SU}}} + \kappa\tau \sum_{k=1}^{K} a_k^{\mathrm{U}} \left(A_{kd_k^*}\rho_k + B_{kd_k^*}\right) f_k^2 + \sum_{k=1}^{K} \frac{p_k D_k \left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{r_k}, \tag{39}$$

$$\text{s.t.} \quad \frac{\kappa \sum_{k=1}^{K} a_k^{\mathrm{S}} \left(A_{kd_k^*}\rho_k + B_{kd_k^*}\right)}{F_{\mathrm{S}}} + \frac{\sum_{k=1}^{K} \left[a_k^{\mathrm{S}}\rho_k D_k + \left(1 - a_k^{\mathrm{S}}\right) D_k\right]}{r_{\mathrm{SU}}} + \frac{d^{\mathrm{SU}}}{c} + \frac{\kappa a_k^{\mathrm{U}} \left(A_{kd_k^*}\rho_k + B_{kd_k^*}\right)}{f_k} + D_k \frac{\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)}{r_k} \leq T, \quad \forall k \in \mathcal{K}, \tag{39a}$$

$$C_{kd_k^*} \leq \rho_k \leq C_{k(d_k^*-1)}, \quad \forall k \in \mathcal{K}, \tag{39b}$$

which is a linear optimization problem and can be addressed using existing toolbox.

Algorithm 1 provides a summary of the semantic compression ratio optimization algorithm.

Algorithm 1 Semantic Compression Ratio Optimization

1: Initialize $\boldsymbol{\rho}$.
2: Rewrite the computation overhead function according to (35).
3: Approximate the value of each segment using (37).
4: Solve problem (38) using dual method, and obtain the optimal segment indicator $\boldsymbol{\alpha}$.
5: Reformulate problem (39) according to the obtained segment indicator $\boldsymbol{\alpha}$.
6: Solve problem (39) using existing toolbox.
7: **Output**: The optimized $\boldsymbol{\rho}$.

C. Optimal Computation Capacity Allocation

With given satellite-UAV computation task allocation, semantic compression ratio, power, and bandwidth allocation, altitude, beamwidth, and location planning, problem (24) can be simplified as

$$\min_{\mathbf{f}} \quad \sum_{k=1}^{K} a_k^{\mathrm{U}} O_k(\rho_k) f_k^2, \tag{40}$$

$$\text{s.t.} \quad t_{\mathrm{S}} + t_{\mathrm{SU}} + \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{f_k} + t_k^{\mathrm{UG}} \leq T, \forall k \in \mathcal{K}, \tag{40a}$$

$$\sum_{k=1}^{K} f_k \leq F_{\mathrm{U}}, \tag{40b}$$

$$f_k \geq 0, \forall k \in \mathcal{K}. \tag{40c}$$

To solve problem (40), we obtain the following theorem.

Theorem 1: The optimal solution of problem (40) is

$$f_k = \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{UG}}}, \forall k \in \mathcal{K}. \tag{41}$$

Proof: Please refer to Appendix A. ■

Theorem 1 demonstrates that the UAV allocates the minimum necessary computation capacity f_k to satisfy latency constraints, thereby optimizing energy consumption. By examining the structure of (41), it is evident that f_k exhibits an increasing trend as the latency term diminishes and the computation overhead $O_k(\rho_k)$ escalates, which is consistent with intuitive expectations.

D. Optimal Power and Bandwidth Allocation

With given satellite-UAV computation task allocation, semantic compression ratio, computation capacity allocation, altitude, beamwidth, and location planning, problem (24) can be simplified as

$$\min_{\mathbf{b},\mathbf{p}} \quad \sum_{k=1}^{K} \frac{p_k D_k \left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_k p_k}{\Theta^2 b_k N_0}\right)}, \tag{42}$$

$$\text{s.t.} \quad \frac{D_k \left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_k p_k}{\Theta^2 b_k N_0}\right)} \leq T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}, \quad \forall k \in \mathcal{K}, \tag{42a}$$

$$\sum_{k=1}^{K} p_k \leq P_{\mathrm{U}}, \tag{42b}$$

$$\sum_{k=1}^{K} b_k \leq B_{\mathrm{U}}, \tag{42c}$$

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$$b_k, p_k \geq 0, \forall k \in \mathcal{K}. \tag{42d}$$

It is hard to solve problem (42) due to the non-convexity of the objective function. Hence, we first try to obtain the optimal condition of problem (42) and we have the following lemma.

Lemma 1: The optimal $(\mathbf{b}^*, \mathbf{p}^*)$ of problem (42) satisfies

$$D_k \frac{\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + 1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}}{b_k^* \log_2\left(1 + \frac{G_0 g_k p_k^*}{\Theta^2 b_k^* N_0}\right)} = T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}. \tag{43}$$

Proof: Please refer to Appendix B. ■

According to lemma 1, we can separate p_k^* with

$$p_k^* = \frac{b_k^*\left(2^{\frac{U_k}{b_k^*}} - 1\right)}{V_k}, \forall k \in \mathcal{K}, \tag{44}$$

where

$$U_k = \frac{D_k\left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}}, \tag{45}$$

and

$$V_k = \frac{G_0 g_k}{\Theta^2 N_0}, \tag{46}$$

are both constants in problem (42).

Then, problem (42) can be reformulated as

$$\min_{\mathbf{b}} \quad \sum_{k=1}^{K} \frac{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}}{V_k} b_k\left(2^{\frac{U_k}{b_k}} - 1\right), \tag{47}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} \frac{b_k\left(2^{\frac{U_k}{b_k}} - 1\right)}{V_k} \leq P_{\mathrm{U}}, \tag{47a}$$

$$\sum_{k=1}^{K} b_k \leq B_{\mathrm{U}}, \tag{47b}$$

$$b_k \geq 0, \quad \forall k \in \mathcal{K}. \tag{47c}$$

To solve problem (47), we have the following theorem.

Theorem 2: Problem (47) is a convex optimization problem.

Proof: Please refer to Appendix C. ■

Following theorem 2, problem (47) can be efficiently solved using existing convex optimization toolbox.

E. Optimal Altitude and Beamwidth

With given satellite-UAV computation task allocation, semantic compression ratio, computation capacity, power, bandwidth allocation, and location planning, problem (24) can be simplified as

$$\min_{H_{\mathrm{U}}, \Theta} \quad \sum_{k=1}^{K} \frac{p_k D_k\left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_0 p_k}{\Theta^2\left(\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|^2 + H_{\mathrm{U}}^2\right) b_k N_0}\right)}, \tag{48}$$

$$\text{s.t.} \quad \frac{D_k\left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_0 p_k}{\Theta^2\left(\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|^2 + H_{\mathrm{U}}^2\right) b_k N_0}\right)} \leq T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}, \forall k \in \mathcal{K}, \tag{48a}$$

$$\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\| \leq H_{\mathrm{U}} \tan\Theta, \forall k \in \mathcal{K}, \tag{48b}$$

$$H_{\mathrm{U}}^{\min} \leq H_{\mathrm{U}} \leq H_{\mathrm{U}}^{\max}, \tag{48c}$$

$$\Theta_{\min} \leq \Theta \leq \Theta_{\max}. \tag{48d}$$

We observe that the objective function of problem (48) and the left hand side of constraint (48a) are both increasing functions in H_{U} with given Θ. Denote H_{U}^* as the optimal value of H_{U} in problem (48), we can claim that

$$H_{\mathrm{U}}^* = \max\left\{H_{\mathrm{U}}^{\min}, \frac{L_{\max}}{\tan\Theta}\right\}, \tag{49}$$

where $L_{\max} = \max_{k\in\mathcal{K}}\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|$. Based on (49), we consider the following two cases.

1) Case 1: If $H_{\mathrm{U}}^* = H_{\mathrm{U}}^{\min}$, problem (48) is equivalent to

$$\min_{\Theta} \quad \Theta, \tag{50}$$

$$\text{s.t.} \quad \frac{D_k\left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_0 p_k}{\Theta^2\left[\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|^2 + \left(H_{\mathrm{U}}^{\min}\right)^2\right] b_k N_0}\right)} \leq T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}, \quad \forall k \in \mathcal{K}, \tag{50a}$$

$$L_{\max} \leq H_{\mathrm{U}}^{\min} \tan\Theta, \tag{50b}$$

$$\Theta_{\min} \leq \Theta \leq \Theta_{\max}. \tag{50c}$$

Obviously, the optimal solution of problem (50) is

$$\Theta^* = \max\left\{\Theta_{\min}, \arctan\frac{L_{\max}}{H_{\mathrm{U}}^{\min}}\right\}, \tag{51}$$

which is the minimal value of Θ satisfying constraints (50b) and (50c). Considering constraint (50a), problem (50) is feasible if and only if

$$\Theta^* \leq \min\left\{\Theta_{\max}, \min_{k\in\mathcal{K}}\sqrt{\frac{G_0 g_0 p_k}{I_k b_k N_0\left(2^{J_k} - 1\right)}}\right\}, \tag{52}$$

where $I_k = \left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|^2 + \left(H_{\mathrm{U}}^{\min}\right)^2$ and $J_k = \frac{D_k\left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k\left(T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}\right)}$. Otherwise, problem (50) has no solution.

2) Case 2: If $H_{\mathrm{U}}^{\min} \geq \frac{L_{\max}}{\tan\Theta^*}$, then the optimal solution of case 1 is the optimal solution of problem (48). Otherwise, $H_{\mathrm{U}}^* = \frac{L_{\max}}{\tan\Theta}$. In this case, problem (48) is equivalent to

$$\min_{\Theta} \quad \sum_{k=1}^{K} \frac{p_k D_k\left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_0 p_k}{\Theta^2\left[\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|^2 + \left(\frac{L_{\max}}{\tan\Theta}\right)^2\right] b_k N_0}\right)}, \tag{53}$$

$$\text{s.t.} \quad \frac{D_k\left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_0 p_k}{\Theta^2\left[\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|^2 + \left(\frac{L_{\max}}{\tan\Theta}\right)^2\right] b_k N_0}\right)} \leq T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}, \quad \forall k \in \mathcal{K}, \tag{53a}$$

$$H_{\mathrm{U}}^{\min} \leq \frac{L_{\max}}{\tan\Theta} \leq H_{\mathrm{U}}^{\max}, \tag{53b}$$

$$\Theta_{\min} \leq \Theta \leq \Theta_{\max}. \tag{53c}$$

It is generally hard to obtain the optimal solution of problem (53) in closed form due to its complicated objective function. Hence, we conduct one-dimensional exhaustive search over $[\Theta_{\min}, \Theta_{\max}]$ to obtain the optimal Θ^*.

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Comparing the optimal solution of the above two cases, the one with lower objective value is the optimal solution of problem (48).

F. Optimal Location Planning

With given satellite-UAV computation task allocation, semantic compression ratio, computation capacity, power, bandwidth allocation, altitude, and beamwidth, problem (24) can be simplified as

$$\min_{\mathbf{L}^{\mathrm{U}}} \quad \sum_{k=1}^{K} \frac{p_k D_k \left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_0 p_k}{\Theta^2\left(\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|^2 + H_{\mathrm{U}}^2\right) b_k N_0}\right)}, \tag{54}$$

$$\text{s.t.} \quad \frac{D_k \left[\left(a_k^{\mathrm{S}} + a_k^{\mathrm{U}}\right)\rho_k + \left(1 - a_k^{\mathrm{S}} - a_k^{\mathrm{U}}\right)\right]}{b_k \log_2\left(1 + \frac{G_0 g_0 p_k}{\Theta^2\left(\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|^2 + H_{\mathrm{U}}^2\right) b_k N_0}\right)} \le T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{U}}, \quad \forall k \in \mathcal{K}, \tag{54a}$$

$$\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\| \le H_{\mathrm{U}} \tan\Theta, \quad \forall k \in \mathcal{K}. \tag{54b}$$

Constraint (54a) is equivalent to

$$\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\| \le \sqrt{\frac{G_0 g_0 p_k}{\Theta^2 b_k N_0 \left(2^{J_k} - 1\right)} - H_{\mathrm{U}}^2}, \quad \forall k \in \mathcal{K}. \tag{55}$$

Denote the right-hand side of (55) by Q_k, then we can combine constraints (54a) and (54b) as

$$\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\| \le \min\left\{H_{\mathrm{U}} \tan\Theta, Q_k\right\}, \quad \forall k \in \mathcal{K}. \tag{56}$$

Since $\left\|\mathbf{L}^{\mathrm{U}} - \mathbf{L}_k^{\mathrm{G}}\right\|$ represents the horizontal distance between the UAV and GT k, the feasible region for GT k is a circular area of radius $\min\left\{H_{\mathrm{U}} \tan\Theta, Q_k\right\}$ with center $\mathbf{L}_k^{\mathrm{G}}$. Denote the feasible region for GT k by $\mathcal{R}_k$, we can express the feasible region of problem (54) as

$$\mathcal{R}_{\mathrm{U}} = \bigcap_{k \in \mathcal{K}} \mathcal{R}_k. \tag{57}$$

Then, we conduct two-dimensional exhaustive search over $\mathcal{R}_{\mathrm{U}}$ to obtain the optimal $\mathbf{L}^{\mathrm{U}}$ with the lowest objective value of problem (54).

G. Algorithm Analysis

The overall SAGIN-enabled PSCom network energy minimization algorithm solve the six subproblems iteratively.

1) Convergence Analysis: Denote the objective value of problem (24) at i-th iteration by $V_{\mathrm{obj}}^{(i)}$, and the objective value at i-th iteration after solving the first subproblem by $V_{\mathrm{s1}}^{(i)}$, etc. We have

$$\begin{aligned} V_{\mathrm{obj}}^{(i-1)} &\ge V_{\mathrm{s1}}^{(i)} \ge V_{\mathrm{s2}}^{(i)} \ge V_{\mathrm{s3}}^{(i)} \\ &\ge V_{\mathrm{s4}}^{(i)} \ge V_{\mathrm{s5}}^{(i)} \ge V_{\mathrm{s6}}^{(i)} = V_{\mathrm{obj}}^{(i)}, \end{aligned} \tag{58}$$

which means the objective value of problem (24) is non-increasing along the iteration. Moreover, the physical meaning of the objective value is energy consumption, which is always positive. Since the objective value is non-increasing during the iteration and is lower-bounded by zero, the proposed iterative algorithm must converge.

TABLE II
MAIN SYSTEM PARAMETERS

Parameter	Value
Number of GTs K	4
Original data size D_k	64 KB
Satellite-UAV distance d^{SU}	200 km
Satellite beam gain δ_{S}	25 dB
Satellite-UAV wavelength λ^{SU}	10 mm
Satellite bandwidth B_{SU}	1 GHz
Satellite transmit power P_{S}	1 W
Power spectral density of AWGN N_0	-174 dBm/Hz
Reference gain g_0	1.42×10^{-4}
Computation coefficient τ	10^{-28}
Satellite computation capacity F_{S}	1 GHz
Maximum latency T	700 ms
Total transmit power of the UAV P_{U}	1 W
$\left[H_{\mathrm{U}}^{\min}, H_{\mathrm{U}}^{\max}\right]$	$[50, 500]$ m
Total bandwidth of the UAV B_{U}	10 MHz
UAV computation capacity F_{U}	0.5 GHz
$\left[\Theta_{\min}, \Theta_{\max}\right]$	$[0, \pi/2]$ rad

2) Complexity Analysis: The complexity of solving problem (24) lies in solving six subproblems at each iteration. For the satellite-UAV computation task allocation subproblem, the complexity is $\mathcal{O}\left(N_1 K\right)$, where N_1 is the number of iterations of using the dual method to solve problem (25). For the semantic compression ratio optimization subproblem, the complexity lies in the segment selection problem and the subsequent convex problem. For the segment selection problem, the complexity is $\mathcal{O}\left(N_2 K D\right)$, where N_2 is the number of iterations of using the dual method to solve problem (38). For the subsequent convex problem, the complexity is $\mathcal{O}\left(M_1^2 M_2\right)$ [43], where $M_1 = K$ is the number of variables, and $M_2 = 2K$ is the number of constraints in problem (39). As a result, the total complexity of solving the semantic compression ratio optimization subproblem is $\mathcal{O}\left(N_2 K D + K^3\right)$. For the optimal computation capacity allocation subproblem, the complexity is $\mathcal{O}\left(K\right)$. For the optimal power and bandwidth allocation subproblem, the complexity is $\mathcal{O}\left(K^3\right)$. For the optimal optimal altitude and beamwidth subproblem, the complexity is $\mathcal{O}\left(\left(\Theta_{\max} - \Theta_{\min}\right)/\eta\right)$, where η is the step size of one-dimensional exhaustive search of problem (53). For the optimal location planning subproblem, the complexity is $\mathcal{O}\left(N_3\right)$, where N_3 is the number of steps of two-dimensional exhaustive search of problem (54). As a result, the total complexity of the proposed algorithm is $\mathcal{O}(N N_1 K + N N_2 K D + N K^3 + N\left(\Theta_{\max} - \Theta_{\min}\right)/\eta + N N_3)$, where N is the number of outer iterations.

IV. SIMULATION RESULTS AND ANALYSIS

In the simulations, the GTs are uniformly distributed within a circular area of radius 300 m. We assume that each GT requires the same amount of data. For the PSCom model, we adopt the same parameters as in [33]. A summary of the main system parameters is provided in Table II.

Fig. 6 demonstrates the convergence behavior of the proposed algorithm under varying satellite computation capacities. The results indicate that the algorithm converges rapidly, requiring only three iterations to achieve stability, which

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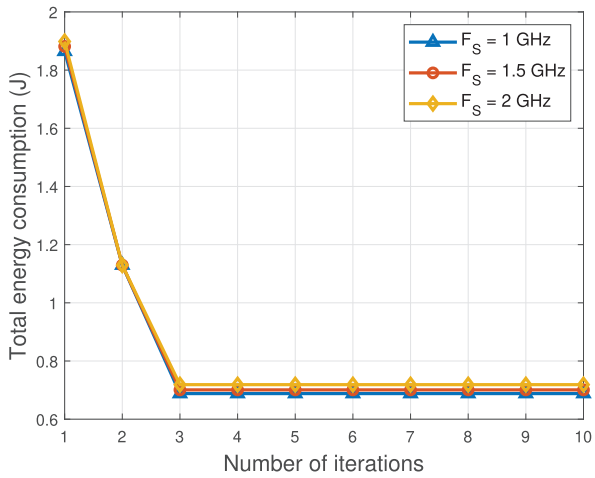

Fig. 6. Convergence behavior of the proposed algorithm.

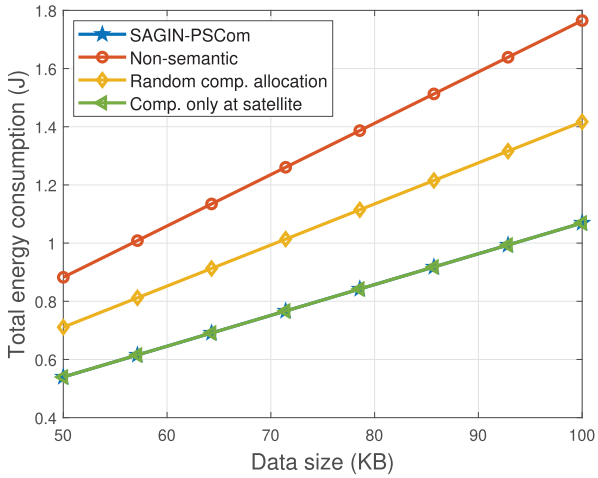

Fig. 7. Total energy consumption vs. data size.

underscores the effectiveness of our optimization algorithm. Initially, the energy consumption is high because communication and computation resources are equally allocated to each GT. However, after several iterations, the energy consumption significantly decreases, as the proposed algorithm effectively optimizes these system parameters.

To compare the results of the proposed algorithm, labeled as 'SAGIN-PSCom', we consider five alternative schemes: the 'Non-semantic' scheme, which employs no semantic compression; the 'Random comp. allocation' scheme, where computation tasks are randomly allocated between the satellite and the UAV; the 'Comp. only at satellite' scheme, which only allocates computation tasks to the satellite; the 'Simplified PSCom' scheme, where conditional probabilities are not used for semantic compression; and the 'Fix UAV location' scheme, which excludes optimization of the UAV's location.

Fig. 7 illustrates the relationship between total energy consumption and the original data size. As expected, increasing the data size results in a corresponding rise in energy consumption across all four examined schemes. Notably, the proposed 'SAGIN-PSCom' scheme consistently achieves the lowest total energy consumption, primarily due to its use of

TABLE III
COMPUTATION TASK ALLOCATION WITH DIFFERENT NUMBERS OF GTS

Number of GTs	1	4	7
Comp. at the satellite	1	4	6
Comp. at the UAV	0	0	1
Total Energy Consumption	0.172 J	0.688 J	1.123 J

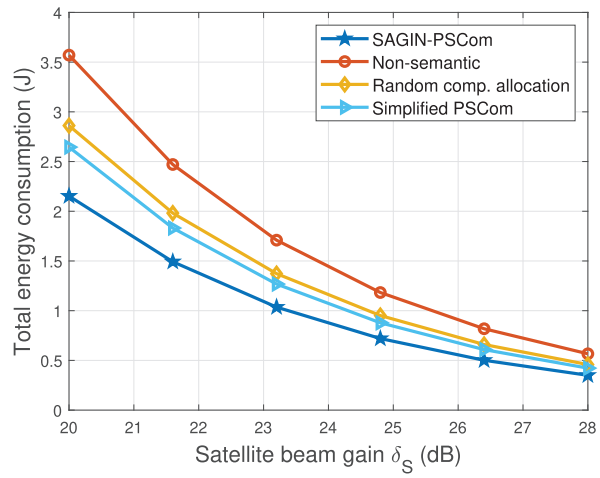

Fig. 8. Total energy consumption vs. satellite beam gain.

the PSCom technique, which effectively compresses the original data and reduces communication energy costs. Although semantic compression incurs additional computation overhead, the energy savings from reduced data transmission outweigh the computational energy expenditure, particularly when an optimized semantic compression ratio is applied. Furthermore, the 'Random comp. allocation' scheme consumes more energy than both the 'Comp. only at satellite' and 'SAGIN-PSCom' schemes, which exhibit identical performance. This suggests that, under the current simulation settings, the 'SAGIN-PSCom' scheme also allocates all computation tasks to the satellite. However, this strategy is not necessarily optimal under all conditions. To further analyze the impact of computation task allocation, Table III examines how the distribution of computation tasks between the satellite and the UAV varies with different numbers of GTs.

In Table III, when the number of GTs is 7, we configure one GT to require less data compared to the others. In this case, the proposed algorithm assigns the computation task of this GT to the UAV, rather than allocating all computation tasks to the satellite. This result demonstrates that the algorithm can dynamically adapt computation task allocation based on the heterogeneous data demands of each GT, ensuring an efficient balance between communication and computation resources.

Fig. 8 demonstrates that the 'SAGIN-PSCom' scheme's benefits are particularly pronounced under conditions of low satellite beam gain. This occurs because a decrease in satellite beam gain diminishes the achievable rate between the satellite and the UAV, thereby increasing the energy required to transmit the same amount of data. Under these conditions, the 'SAGIN-PSCom' scheme compensates by allocating more

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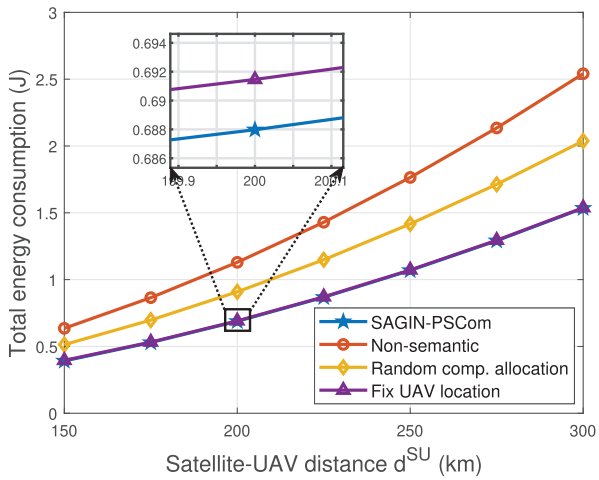

Fig. 9. Total energy consumption vs. satellite-UAV distance.

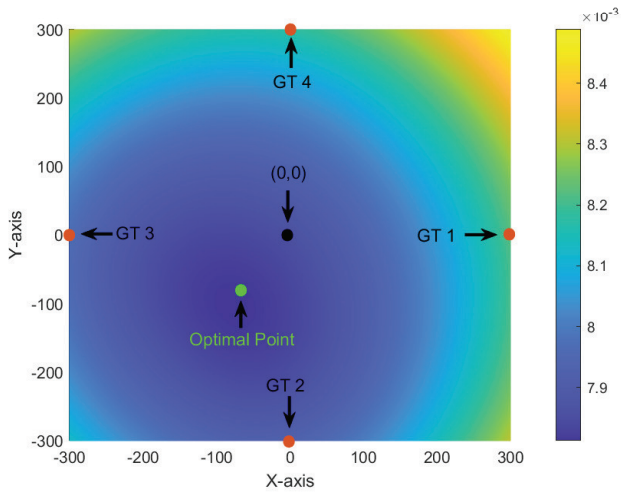

Fig. 10. The impact of UAV's location on UAV communication energy.

computation resources at the satellite to mitigate the adverse effects of reduced beam gain. Furthermore, the performance of the 'Simplified PSCom' scheme falls between that of the 'Non-semantic' and 'SAGIN-PSCom' schemes, highlighting the advantage of leveraging deeper probabilistic information.

Fig. 9 depicts the relationship between total energy consumption and the distance between the satellite and UAV. Interestingly, the 'Fix UAV location' scheme exhibits energy consumption levels comparable to those of the SAGIN-PSCom' scheme, suggesting that the UAV's location does not critically impact the system's overall energy efficiency. This observation also indicates that the energy used for communication between the UAV and GTs is relatively minor within the system's total energy consumption.

In Fig. 10, the influence of UAV's location on its communication energy is explored. The axes represent the UAV's two-dimensional coordinates in meters, and the color gradient illustrates varying values of the objective function defined in Section III-F, which is the communication energy consumed by the UAV. Fig. 10 is obtained when the 4 GTs requiring different amount of data. The figure reveals that the optimal UAV location is not at the origin but slightly offset. Given that the UAV communication energy is quantified on the order of 10^{-3}, it constitutes a minor fraction of the total system energy consumption. This minor impact supports the observation that the 'Fix UAV location' scheme performs comparably to the 'SAGIN-PSCom' scheme.

V. Conclusion

This paper has investigated the problem of energy efficiency in SAGIN-enabled PSCom system. The model considers that a satellite transmits data to multiple GTs through a UAV acting as a relay. The satellite and the UAV can use PSCom technique to compress the transmitted data, while the GTs can automatically recover missing information. The PSCom is enabled by shared probabilistic graphs among the transceivers, allowing for the conservation of communication resource at the expense of additional computation resource. The joint communication and computation problem is formulated as an optimization problem aiming to minimize the total communication and computation energy consumption of the network under latency, power, computation capacity, bandwidth, semantic compression ratio, and UAV location constraints. We proposed an iterative algorithm to solve this non-convex non-smooth problem, where the closed-form solutions for computation capacity allocation and UAV altitude are obtained at each iteration. Numerical results demonstrate the effectiveness of the proposed algorithm.

In future research, we plan to expand our scenario to include multiple satellites and UAVs. In addition, we plan to consider the dynamic elements of satellite motion and the trajectory of the UAV to improve the robustness and adaptability of our energy-efficient communication framework.

Appendix A
Proof of Theorem 1

For those GTs with $a_k^{\mathrm{U}} = 0$, we can simply set $f_k = 0$ because the UAV does not need to compute for these GTs. For other GTs with $a_k^{\mathrm{U}} = 1$, we can combine constraints (40a) and (40c) as

$$f_k \geq \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{UG}}} > 0, \forall k \in \mathcal{K}, \tag{59}$$

where $\frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{UG}}}$ is a constant in problem (40).

Then, the Lagrange function of problem (40) can be given by

$$\begin{aligned} L(\mathbf{f}, \boldsymbol{\mu}_1, \mu_2) = &\sum_{k=1}^{K} a_k^{\mathrm{U}} O_k(\rho_k) f_k^2 \\ &- \sum_{k=1}^{K} \mu_{1k} \left(f_k - \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{UG}}} \right) \\ &+ \mu_2 \left(\sum_{k=1}^{K} f_k - F_{\mathrm{U}} \right), \end{aligned} \tag{60}$$

where $\boldsymbol{\mu}_1 = [\mu_{11}, \cdots, \mu_{1K}]$ is the non-negative Lagrange multiplier vector associated with constraint (59), and μ_2 is the

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non-negative Lagrange multiplier associated with constraint (40b). The first derivative of (60) is

$$\frac{\partial L\left(\mathbf{f}, \boldsymbol{\mu}_1, \mu_2\right)}{\partial f_k} = 2a_k^{\mathrm{U}} O_k(\rho_k) f_k - \mu_{1k} + \mu_2. \tag{61}$$

Setting $\frac{\partial L(\mathbf{f},\boldsymbol{\mu}_1,\mu_2)}{\partial f_k} = 0$ yields

$$f_k = \frac{\mu_{1k} - \mu_2}{2a_k^{\mathrm{U}} O_k(\rho_k)}. \tag{62}$$

According to complementary slackness, we have

$$\mu_{1k}\left(f_k - \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{UG}}}\right) = 0. \tag{63}$$

To obtain the Karush-Kuhn-Tucker (KKT) point, conditions (62) and (63) must be satisfied at the same time. Moreover, due to the fact that $f_k > 0$ and μ_{1k}, μ_2 are non-negative, μ_{1k} must be greater than zero, which means

$$f_k - \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{UG}}} = 0, \tag{64}$$

for those GTs with $a_k^{\mathrm{U}} = 1$. As mentioned above, we set $f_k = 0$ for those GTs with $a_k^{\mathrm{U}} = 0$. Hence, we can obtain the closed-form solution of problem (40) as

$$f_k = \frac{\kappa a_k^{\mathrm{U}} O_k(\rho_k)}{T - t_{\mathrm{S}} - t_{\mathrm{SU}} - t_k^{\mathrm{UG}}}, \forall k \in \mathcal{K}. \tag{65}$$

Appendix B
Proof of Lemma 1

Lemma 1 can be proved by the contradiction method. Define function

$$f(x) = \frac{x}{\log_2(1+x)}, x > 0, \tag{66}$$

whose derivative is

$$f'(x) = \frac{(\ln 2)\,(1+x)\log_2(1+x) - x}{(\ln 2)\left[\log_2(1+x)\right]^2 (1+x)}, x > 0. \tag{67}$$

Then, define function

$$g(x) = (\ln 2)\,(1+x)\log_2(1+x) - x, x > 0, \tag{68}$$

whose derivative is

$$g'(x) = (\ln 2)\log_2(1+x), x > 0. \tag{69}$$

Obviously, $g'(x) > 0$ for $x > 0$. Since $g(0) = 0$, we have $g(x) > 0$ for $x > 0$. Furthermore, since the denominator of $f'(x)$ is greater than zero for $x > 0$, we have $f'(x) > 0$ for $x > 0$. Thus, $f(x)$ is monotonically increasing on $x > 0$. Hence, with given b_k, the objective function of problem (42) increases with growing p_k.

Assume $(\mathbf{b}, \mathbf{p})$ is a feasible solution of problem (42), if there exists one k whose corresponding constraint (42a) holds with inequality, we can always decrease p_k to obtain a smaller objective value. Therefore, for optimal $(\mathbf{b}^*, \mathbf{p}^*)$, constraint (42a) must hold with equality.

Appendix C
Proof of Theorem 2

Define function

$$q(x) = x\left(2^{\frac{a}{x}} - 1\right), x > 0, \tag{70}$$

where a is a positive constant. Then, we can write its derivative as

$$q'(x) = 2^{\frac{a}{x}} - 1 - \frac{a(\ln 2)2^{\frac{a}{x}}}{x}, x > 0. \tag{71}$$

Furthermore,

$$q''(x) = \frac{a^2(\ln 2)^2 2^{\frac{a}{x}}}{x^3}, x > 0. \tag{72}$$

Obviously, $q''(x) > 0$ on $x > 0$. Thus, $q(x)$ is a convex function. Therefore, the objective function of problem (47) and constraint (47a) are both convex. Since constraints (47b) and (47c) are also convex, problem (47) is a convex optimization problem.

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